

Programme	B. Sc. STATISTICS				
Course Code	STA 8 CJ 407				
Course Title	APPLIED MULTIVARIATE TECHNIQUES				
Type of Course	Major				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Good knowledge of Multivariate Normal distribution. 2. Programming skill using R.				
Course Summary	The main objective of this course are to : 1. Inculcate deep knowledge on various multivariate techniques. 2. Develop clear idea on when and where to use dependence and interdependence multivariate methods. 3. Bridge the relation between multivariate analysis using software, to strengthen statistical applications in diversified spectrum of life.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Analysing Multivariate data using reduction techniques like Principal Component Analysis, Factor Analysis and Canonical correlation.	An	P	Seminar Presentation /
2.	Formulate multivariate hypothesis tests and drawing appropriate conclusions	C	M	Instructor-created exams
3.	Discriminate multivariate normal population.	E	C	Group Tutorial Work
4.	Identify data reduction techniques	U	C	Home Assignments
5.	Analyse multivariate data using statistical software's.	An	P	Practical Skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Principle Component- Factor Analysis-Canonical correlation		16	20
	1	Principle component		
	2	Maximum likelihood estimates of the principal components and their variance		
	3	Extraction of Principal Components and their variances		
	4	Factor Analysis – Mathematical model –Estimation of Factor Loading		
	5	Canonical correlation – Estimation of canonical correlation and variates		
	6	Structural equation models.		
II	Classification Problems		15	20
	7	Classification problems		
	8	Classification into one of two population (known and unknown dispersion matrix)		
	9	Classification in to one of several populations		
	10	Multivariate analysis of variance (MANOVA) – One way and two way classification		
	11	Tests independence of sets of variables		
	12	Equality of dispersion matrices and Sphericity test.		
III	Discriminant Analysis		9	15
	13	Discriminant Analysis		
	14	Likelihood ratio method		
	15	Bayes and min-max procedure		
	16	Discrimination between two multivariate normal population with common dispersion		
	17	Sample discriminate function		
	18	Estimation – Fisher’s method for discriminating among several populations.		
IV	Cluster Analysis		8	15
	19	Cluster Analysis		
	20	Proximity measures		
	21	Hierarchical clustering techniques : single, complete and average linkage algorithms.		
	22	Non-hierarchical clustering techniques : K means method.		
V	Open –End		12	
	Problems regarding Module I to IV using Statistical software			

Text Book

1. **Anderson T W (2010)** : An Introduction to Multivariate Statistical Analysis, Wiley Eastern Ltd.
2. **Johnson, R A and Wichern D W (2003)** : Applied Multivariate Statistical Analysis, Prentice-Hall of India Private Ltd., New Delhi.

Reference

1. **Morrison F (2003)** : Multivariate Statistical Methods, Brooks/Cole, 4th Revised edn., McGraw Hill Book Company
2. **Seber G A (2004)** : Multivariate Observations, John Wiley.
3. **Denis, D J (2021)** : Applied Univariate, Bivariate and Multivariate Statistics : Understanding Statistics for Social and Natural Scientists, With Application in SPSS and R, John Wiley & Sons.